



SIGN-OFF REQUIRED

This assessment must be digitally signed off by an Authorised User.

General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	02/06/2025
Signed Off By	This assessment has not yet been signed off
Signed Off On	--
Sign Off Comments	
Operation Assessed	Demonstration of Humanoid Robot

Associated with specific area

-  Basement 2
-  B2 029 (Robotics Lab)
-  CTP

Description of work area and/or activity assessed

This RA (Risk Assessment) pertains to the demonstration activities of Humanoid Robots intended for Community Engagement and Promotional Activities at both internal (NYUAD) and external public events, such as Learning Festivals, Competitions, Hackathons, Conferences, Celebrations, Exhibitions, Shows and more.

Reason for Risk Assessment

To predict and avoid any potential risks during the operation of the robot.

Overall Current Risk

Moderate Risk



Average Number of Persons Affected

2	 Children
2	 Employees
2	 Machine Operators
2	 Members of the Public
2	 Visitors

Current Rating	Hazard Information	Task Status
Moderate Risk	Moving Parts - Loss of stability and falls The robot stands up and keeps balancing only with active control. However, active control does not ensure that stable balancing is always attained. - Pinch points during motion Pinch points are present around the joints (see Pinch points) and cannot be entirely eliminated without compromising the mobility of the limbs, or the ability to walk and maintain balance. The robot pinch points are situated in areas that are not normally reached in situations other than handling and transportation (while the robot is powered off). When motors are active, there is no foreseeable use that would require access to the limb joints. - Hazards associated with stopping or other powerless motion The robot will automatically disable motor power in the following cases: • As a result of a detected failure or a detected fall. • As a result of a safety-related stopping function forcing a de-energized state.	0 0 1
Measures Currently in place to prevent risk of injury		
Software Emergency switch (Soft) Sensors to proximity from humans and objects		

Potential Rating	Additional Controls Required		
Low Risk	Additional Controls and Actions Operators should be trained in the use of the robots. Operators should experienced in the use of the robots. Operators will not navigate at high speeds Operators will constantly remind the public to not approach the robots. Operators will monitor the robots for signs of abnormal operation.	Due by / Whom 02/06/2025 S.Prieto Ayllon	Action Status Complete

Current Rating	Hazard Information	Task Status
Low Risk 	Fire - Fire, explosion and hazardous materials There are no known significant risks associated with the robot's lubricants.	0 0 1
	Measures Currently in place to prevent risk of injury	
	Systems monitoring the status of the robot's operation and alert the user if a function or a component are not working properly or if the measurement values (Temperature, Voltage, Current, etc) are out of range. If there is a problem on the crucial systems of the robots the will "lock" and cannot not operate.	

Potential Rating	Additional Controls Required		
Low Risk 	Additional Controls and Actions		Due by / Whom
			Action Status
	Operator user training	02/06/2025 S.Prieto Ayllon	Complete

Current Rating	Hazard Information	Task Status
Low Risk 	Electricity The robot uses a separated extra low-voltage (SELV) power supply design (<60 VDC), meeting the requirements in IEC 60204-1 for electrical equipment and IEC 62133 for batteries. There are no known significant residual risks associated with shocks, either apparent or as a result of faults.	0 0 1
	Measures Currently in place to prevent risk of injury	
	IEC 60204-1 for electrical equipment and IEC 62133 for batteries.	

Potential Rating	Additional Controls Required		
Low Risk 	Additional Controls and Actions		Due by / Whom
			Action Status
	Operator user training	02/06/2025 S.Prieto Ayllon	Complete

Related Files